Lab 4 – Endpoint Detection and Response, and Incident Investigation.

Send your report only to <csclass.dz@gmail.com>
Team information (one report by team)
- Boulgamhe Lahcen Nidhal

Deadline: Monday, Apr 13, 2026

Lab objective

Simulate attacks on endpoints and investigate them using EDR and SIEM tools. The goal is to rely on **hands-on cyber-range and incident-response exercises** to learn real-world skills rather than purely theoretical security courses.

Modern SOC teams rarely rely on only one tool. Instead, they combine
- **EDR investigation** (Velociraptor, CrowdStrike, Defender)
- **Security monitoring** (SIEM like Wazuh)
- **Attack simulation** (Kali or red-team tools)

The lab combines
- **EDR investigation:** Velociraptor
- **Security visibility/logging:** Wazuh
- **Attack simulation:** Kali Linux
  This structure mirrors how **SOC environments combine endpoint telemetry with investigation tools**.

In this lab, you will
- Understand how **EDR tools monitor endpoint activity**
- Perform **endpoint investigation using Velociraptor**
- Observe alerts generated by Wazuh
- Investigate suspicious system artifacts
- Reconstruct attacker activity on compromised machines
- Map attacker behavior to the MITRE ATT&CK framework
- Produce a **SOC incident investigation report**

Developed Skills

After completing this lab, you will be able to
- Investigate endpoint incidents using **EDR tools**
- Analyze endpoint telemetry

- Identify attacker techniques
- Use Velociraptor artifacts for forensics
- Correlate alerts and endpoint evidence
- Produce a **SOC investigation report**

Required Environment

Deploy a small **cyber-range environment**

Machine	Role
Kali Linux	Attacker
Windows VM	Victim endpoint
Linux VM	Victim endpoint
Velociraptor server	Endpoint investigation
Wazuh server	Security monitoring

Agents installed

- Velociraptor agent: Windows and Linux
- Wazuh agent: Windows and Linux

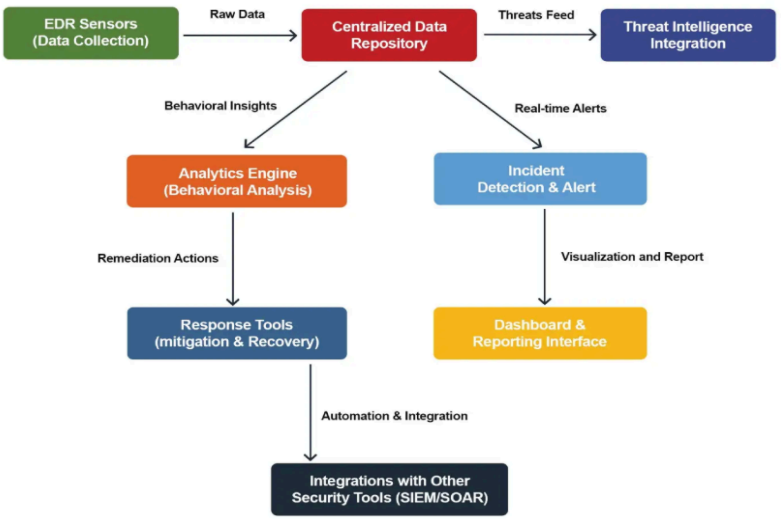
General Questions

1/ EDR Architecture

- Provide the general architecture of an EDR solution. Include the role of the agent, telemetry collection, server/console, and alert generation.

Question 1 — EDR Architecture

1.1 General Architecture of an EDR Solution



1. EDR Sensors

EDR sensors are lightweight agents placed on endpoints like computers, servers, and mobile devices. They monitor what's happening on these devices, such as modifications, network connections, process execution and termination, and user behavior. These sensors send telemetry data to a central repository for analysis.

2. Centralized Data Repository

The centralized data repository is a safe place for keeping telemetry data gathered by sensors. It stores records of endpoint activities, which helps in understanding past events for forensic investigations and rebuilding the timeline of attacks.

3. Threat Intelligence Integration

Threat intelligence integration enhances EDR capabilities by using global threat data to identify known threats and Indicators of Compromise (IOCs). This feature compares endpoint activities with updated threat intelligence databases, offering real-time updates on new threats

4. Analytics Engine

The analytics system examines data gathered from various sources using sophisticated methods like machine learning, behavior analysis, and anomaly detection. It identifies unusual activity and Zero-Day threats, enabling more effective threat detection.

5. Incident Detection and Alerts

The incident detection and alerts component continuously monitor endpoint data for potential security incidents. It generates real-time alerts and notifies the Security Operations Center (SOC) when threats are identified.

6. Response Tools

Response tools help in swiftly containing and mitigating threats. They can do things like isolating infected devices from the rest, terminate malicious processes, and quarantine suspicious files to keep them from being accessed.

7. Dashboard and Reporting Interface

The dashboard and reporting interface offers a simple platform for security teams to keep track of endpoint actions, analyze incidents, and generate compliance reports.

8. Integration with Other Security Tools

Integration with other security tools, like SIEM and SOAR platforms, expands EDR's abilities by allowing for automation and better threat visibility. This feature improves overall security operations by automating incident responses, streamlining information sharing, and increasing the effectiveness of the security system.

- Describe how Velociraptor implements this architecture, highlighting how endpoint data is collected, processed, and presented to the SOC analyst.
  Goal: Ensure that you understand how EDR works (in particular, Velociraptor), not simply following attack simulation steps.

1.2 How Velociraptor Implements This Architecture

Velociraptor is an open-source EDR and digital forensics platform that implements the above architecture in the following way:

Agent (Client)

The Velociraptor client binary (velociraptor -c client.config.yaml client) is deployed on each endpoint. It implements the VQL (Velociraptor Query Language) execution engine, which means the agent itself can run complex forensic queries locally. The client communicates with the server over HTTPS using a unique client certificate for mutual TLS authentication. The agent registers itself with the server and awaits task assignments (artifact collection requests).

Telemetry Collection via VQL Artifacts

Instead of passively streaming all events, Velociraptor uses a pull/push hybrid model. The analyst selects Artifacts (predefined or custom VQL queries) and deploys them as "hunts" (against all clients) or "flows" (against a specific client). Common artifacts include Windows.System.Pslist (process list), Windows.EventLogs.EvtxHunter (event log search), Windows.Network.Netstat (active connections), and Linux.Sys.BashHistory (bash history). The agent executes the VQL locally and streams results back to the server.

Server and Console

The Velociraptor server (velociraptor -c server.config.yaml frontend) provides a web GUI accessible by analysts. It maintains a datastore of all collected artifacts, a client inventory, and a hunt manager. Analysts can search across collected data, create new hunts, and monitor agent health. The server also supports notebook-style analysis sessions combining VQL queries and markdown annotations.

Alert Integration

While Velociraptor is primarily an investigation tool rather than a real-time alerting SIEM, it integrates with Wazuh (and other SIEMs) by forwarding events or by having Wazuh monitor the Velociraptor server logs. In this lab, Wazuh provides real-time alerting while Velociraptor provides deep forensic investigation capability — a combination that mirrors production SOC environments.

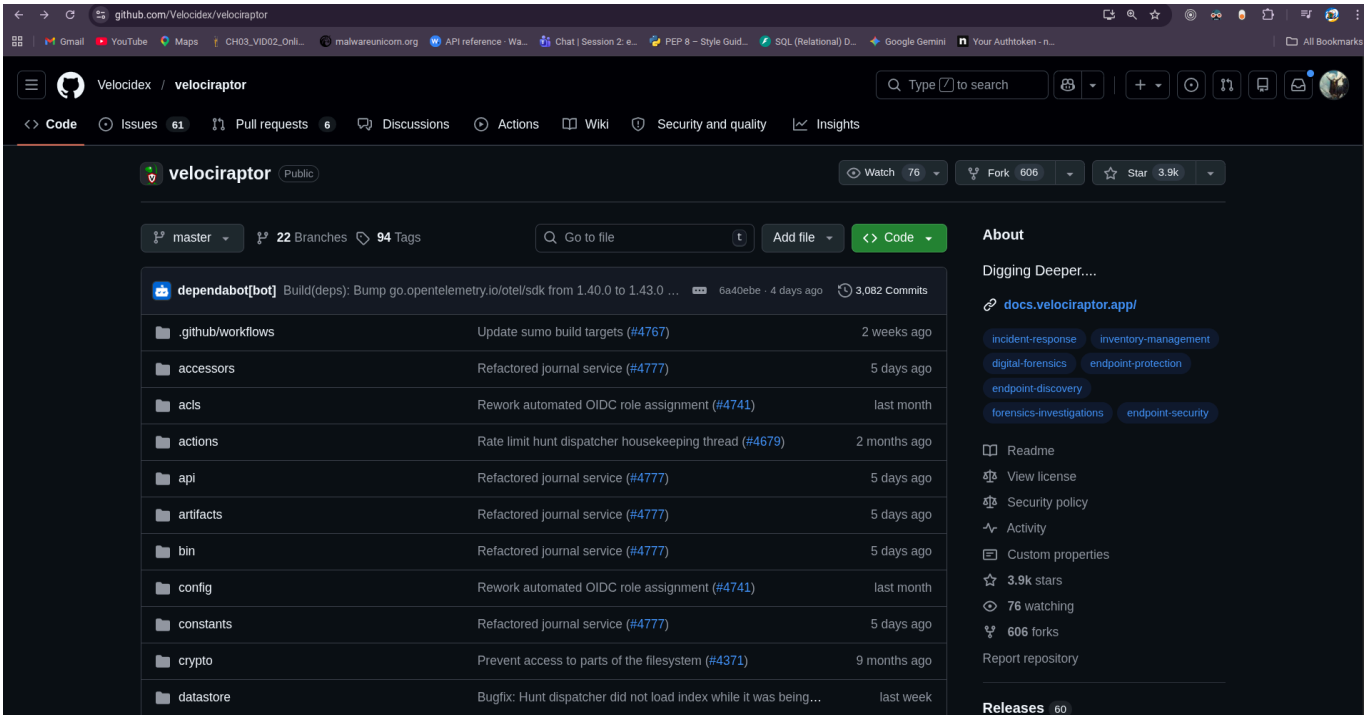
2/ Velociraptor Installation

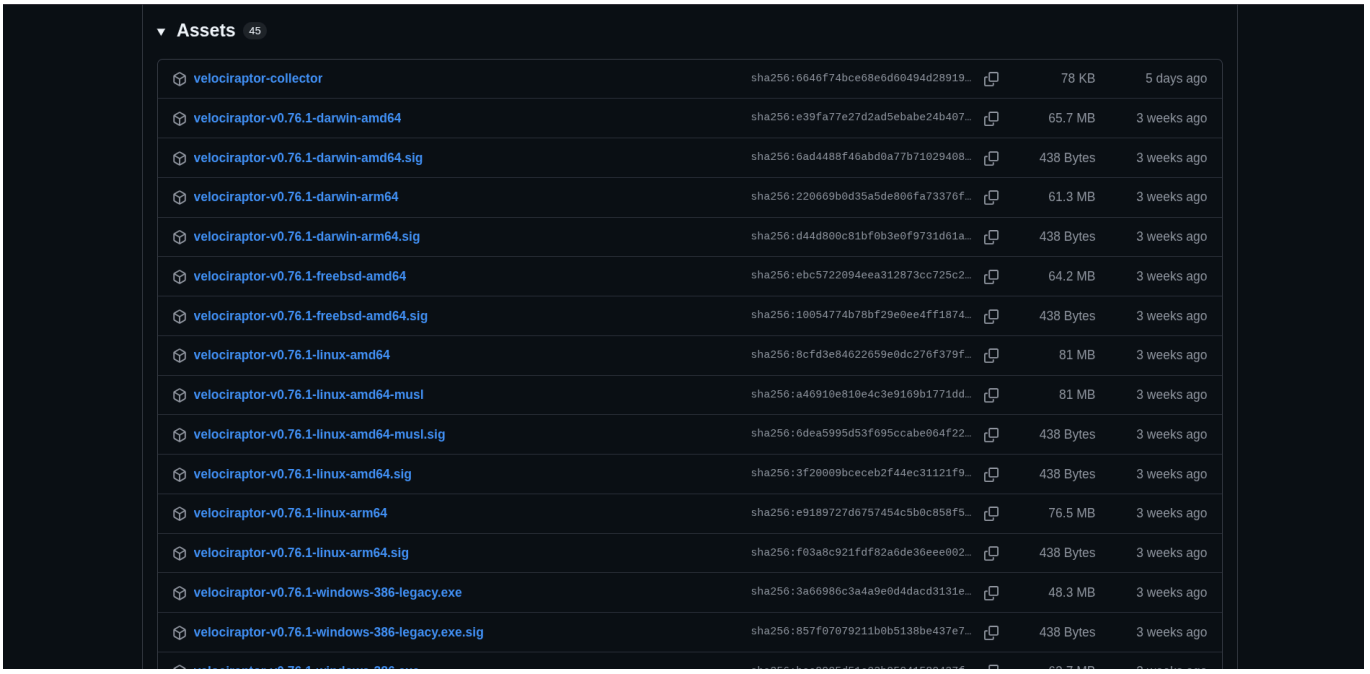
- Explain the steps to install and configure Velociraptor on both Windows and Linux endpoints.

Question 2 — Velociraptor Installation

2.1 Installing and Configuring the Velociraptor Server on RHEL (192.168.8.104)

Step 1 — Download the Velociraptor binary





Step 2 — Generate server configuration

```
[ababil@localhost velociraptor]$ sudo ./velociraptor-v0.76.1-linux-amd64  config generate -i
```

```
 Welcome to the Velociraptor configuration generator

 This wizard creates a configuration file for a new deployment.

 Let's begin by configuring the server itself.


 Deployment Type
 This wizard can create the following distinct deployment types.
   ● Self Signed SSL
     Automatically provision certificates with Lets Encrypt
     Authenticate users with SSO

 What OS will the server be deployed on?
 For production use we recommend to deploy on Ubuntu Linux, but you can deploy on other
↑ up • ↓ down • / filter • enter select
```

```
The datastore directory is where Velociraptor will store all files.
This should be located on a partitian large enough to contain all data you are likely to
collect.
Make sure there is sufficient disk space available!
> /opt/velociraptor

Path to the logs directory.
Velociraptor will write logs to this directory. By default it resides within the datastore
directory but you can place it anywhere.
> logs

Internal PKI Certificate Expiration
By default internal certificates are issued for 1 year.

If you expect this deployment to exist past one year you might
consider extending the default validation.
   1 Year
   2 Years
   10 Years

Use registry for client writeback?
Traditionally Velociraptor uses files to store client state on all operating systems.
shift+tab back • enter next
```

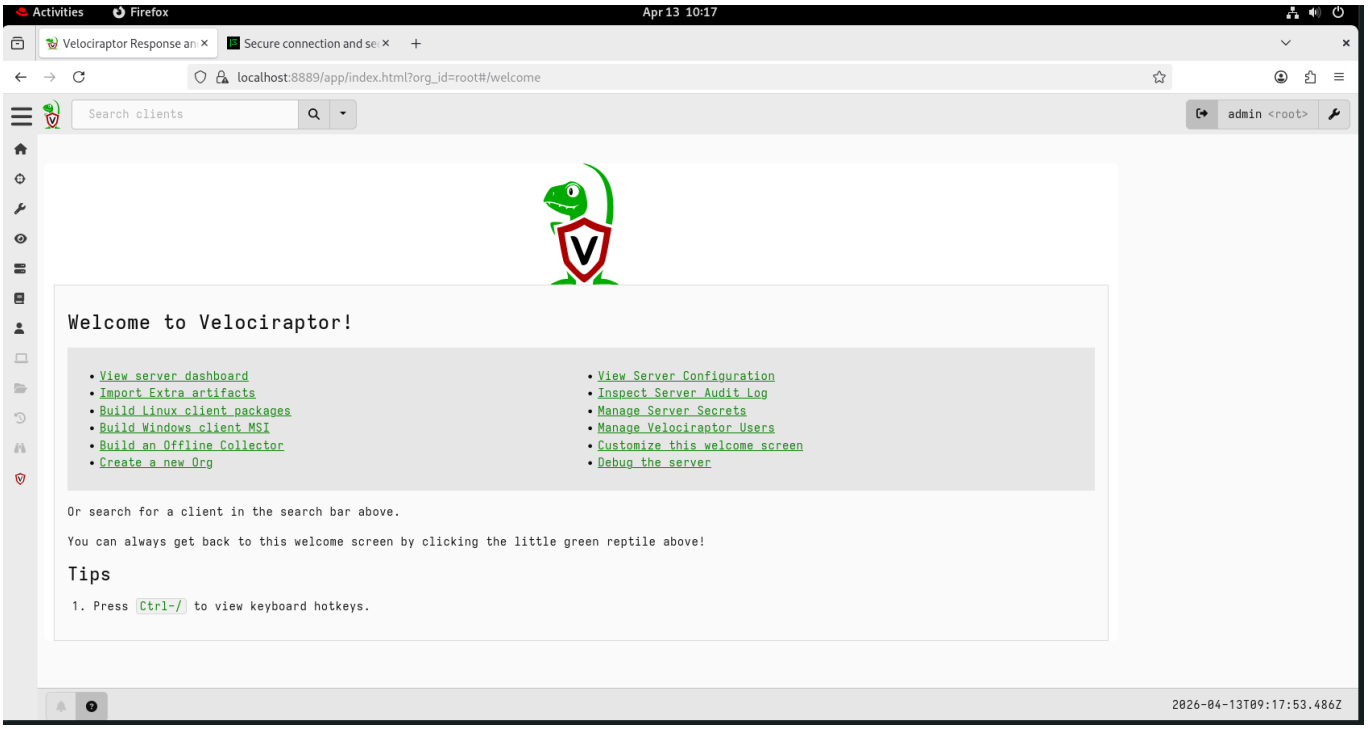
```
[ababil@localhost ~]$ sudo mkdir /opt/velociraptor
[sudo] password for ababil:
[ababil@localhost ~]$ ls /opt/velociraptor
[ababil@localhost ~]$ sudo chown -R ababil /opt/velocir
aptor
[ababil@localhost ~]$
```

```
Velociraptor will write logs to this directory. By default it resides within the datastore
directory but you can place it anywhere.
> /opt/velociraptor/logs

Internal PKI Certificate Expiration
By default internal certificates are issued for 1 year.

If you expect this deployment to exist past one year you might
consider extending the default validation.
   1 Year
   2 Years
   10 Years

Use registry for client writeback?
Traditionally Velociraptor uses files to store client state on all operating systems.

You can instead use the registry on Windows. NOTE: It is your responsibility to ensure the
registry keys used are properly secured!

By default we use HKLM\SOFTWARE\Velocidex\Velociraptor

←/→ toggle • shift+tab back • enter submit • y Yes • n No
```

```
> localhost

DNS Type
In order for the server to be reachable from the internet, you must have DNS configured.
   None - Configure DNS manually
   NOIP
   CloudFlare

Would you like to try the new experimental websocket comms?
Websocket is a bidirectional low latency communication protocol supported by
most modern proxies and load balancers. This method is more efficient and
portable than plain HTTP. Be sure to test this in your environment.

                                   Yes      No

Enter the frontend port to listen on.
> 8000

Enter the port for the GUI to listen on.
> 8889
```

```
version:
  name: velociraptor
  version: 0.76.1
  commit: 3e98857b5
  build_time: "2026-03-25T16:38:27Z"
  compiler: go1.25.3
  system: linux
  architecture: amd64
Client:
  server_urls:
  - https://localhost:8000/
  ca_certificate: |
    -----BEGIN CERTIFICATE-----
    MIIDTDCCAjSgAwIBAgIRAPorBwo+FJvs1av3c7hmSFQwDQYJKoZIhvcNAQELBQAw
    GjEYMBYGA1UEChMPVmVsb2NpcmFwdG9yIENBMB4XDTI2MDQxMzA4NDAwMFoXDTM2
    MDQxMDA4NDAwMFowGjEYMBYGA1UEChMPVmVsb2NpcmFwdG9yIENBMIIBIjANBgkq
    hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAse5/ba2BHt05hEIPm3S3eoBpF1l8DowE
    gM3EcC9Dk2vI7ogu87E21PnAE7KSN85ySFGxswToon4zQk/hYq0nMeMfPkcQkfya
    Kizv6TDCg2hfF4OO15bwE4bw7oybVlAKFkc+O5xHE88MUK73/CLvf7bcPYcWQmtJ
    QrtZAdlJHaVNO9ipNvLwJSU6OEObXXmJqUyFz7kOtpcnqQFgCo5rVTrgw+f6ZaT/
    XDrH8KZjbkA0H95/9OkKODxr4wH2J/nnY4szUDf4k5jqWrbuawXfY7rXdw1tzwVf
    +u5twj/2bwm4z83Cpdu945/0fNEX14nt3bekidtgRCMPlbolfltgaQIDAQABo4GM
                                                                    1,1
```

```
[ababil@localhost velociraptor]$ ./velociraptor-v0.76.1-linux-amd64 --config server.config.yam
l gui
velociraptor-v0.76.1-linux-amd64: error: gui: open server.config.yaml: permission denied
[ababil@localhost velociraptor]$ sudo !!
sudo ./velociraptor-v0.76.1-linux-amd64 --config server.config.yaml gui
[sudo] password for ababil:
[INFO] 2026-04-13T09:14:00Z  _     __      __            _                    __
[INFO] 2026-04-13T09:14:00Z | |   / /__   / /___  _____(_)_______ _____  / /_____  _____
[INFO] 2026-04-13T09:14:00Z | | / / _ \/ / __ \/ ___/ / ___/ __ `/ __ \/ __/ __ \/ ___/
[INFO] 2026-04-13T09:14:00Z | |/ /  __/ / /_/ / /__/ / /__/ /_/ / /_/ / /_/ /_/ / /
[INFO] 2026-04-13T09:14:00Z |___/\___/_/\____/\___/_/_/   \__,_/ .___/\__/\____/_/
[INFO] 2026-04-13T09:14:00Z                                   /_/
[INFO] 2026-04-13T09:14:00Z Digging deeper!                  https://www.velocidex.com
[INFO] 2026-04-13T09:14:00Z This is Velociraptor 0.76.1 built on 2026-03-25T16:38:27Z (3e98857
b5)
[INFO] 2026-04-13T09:14:00Z Loading config from file server.config.yaml
[INFO] 2026-04-13T09:14:00Z Initializing logging for /opt/velociraptor/logs/Velociraptor
```



Step 3 — Create an admin user

I already do it when i configure my server

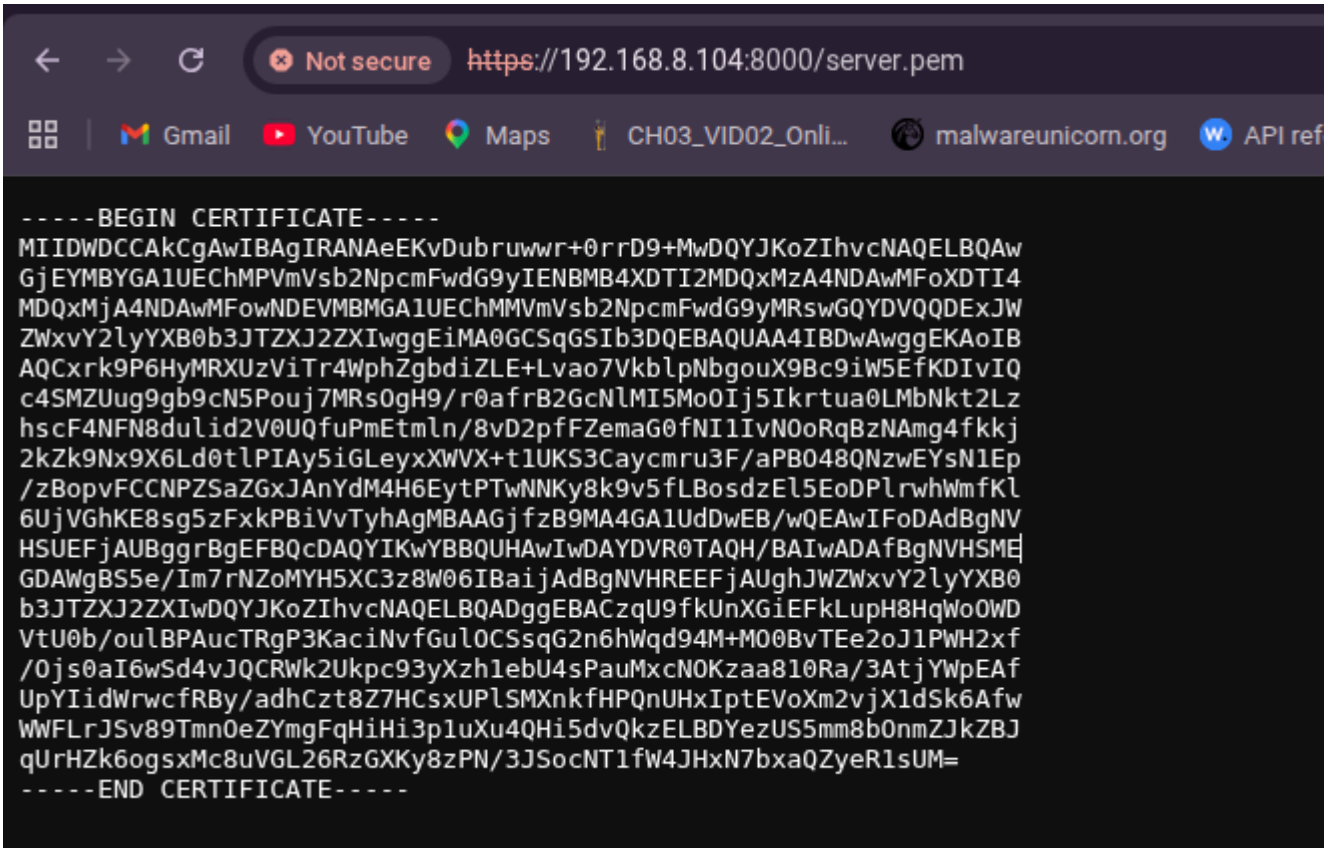
Username = admin , Password = admin

Step 4 — Run as a systemd service

I run as exutable programe but you can run it as demon or service

The Velociraptor web console is then accessible at: https://192.168.8.104:8889

Male sure you run the veloceraoter in frotendmode not GUI mode do can the agent reach his cert at




```
-----BEGIN CERTIFICATE-----
MIIDWDCCAkCgAwIBAgIRANAeEKvDubruwwr+0rrD9+MwDQYJKoZIhvcNAQELBQAw
GjEYMBYGA1UEChMPVmVsb2NpcmFwdG9yIENBMB4XDTI2MDQxMzA4NDAwMFoXDTI4
MDQxMjA4NDAwMFowNDEVMBMGA1UEChMMVmVsb2NpcmFwdG9yMRswGQYDVQQDExJW
ZWxvY2lyYXB0b3JTZXJ2ZXIwggEiMA0GCSqGSIb3DQEBAQUAA4IBDwAwggEKAoIB
AQCxrk9P6HyMRXUzViTr4WphZgbdiZLE+Lvao7VkblpNbgouX9Bc9iW5EfKDIvIQ
c4SMZUug9gb9cN5Pouj7MRsOgH9/r0afrB2GcNlMI5MoOIj5Ikrtua0LMbNkt2Lz
hscF4NFN8dulid2V0UQfuPmEtmln/8vD2pfFZemaG0fNI1IvNOoRqBzNAmg4fkkj
2kZk9Nx9X6Ld0tlPIAy5iGLeyxXWVX+t1UKS3Caycmru3F/aPBO48QNzwEYsN1Ep
/zBopvFCCNPZSaZGxJAnYdM4H6EytPTwNNKy8k9v5fLBosdzEl5EoDPlrwhWmfKl
6UjVGhKE8sg5zFxkPBiVvTyhAgMBAAGjfzB9MA4GA1UdDwEB/wQEAwIFoDAdBgNV
HSUEFjAUBggrBgEFBQcDAQYIKwYBBQUHAwIwDAYDVR0TAQH/BAIwADAfBgNVHSME
GDAWgBS5e/Im7rNZoMYH5XC3z8W06IBaijAdBgNVHREEFjAUghJWZWxvY2lyYXB0
b3JTZXJ2ZXIwDQYJKoZIhvcNAQELBQADggEBACzqU9fkUnXGiEFkLupH8HqWoOWD
VtU0b/oulBPAucTRgP3KaciNvfGulOCSsqG2n6hWqd94M+MO0BvTEe2oJ1PWH2xf
/Ojs0aI6wSd4vJQCRWk2Ukpc93yXzh1ebU4sPauMxcNOKzaa810Ra/3AtjYWpEAf
UpYIidWrwcfRBy/adhCzt8Z7HCsxUPlSMXnkfHPQnUHxIptEVoXm2vjX1dSk6Afw
WWFLrJSv89TmnOeZYmgFqHiHi3p1uXu4QHi5dvQkzELBDYezUS5mm8bOnmZJkZBJ
qUrHZk6ogsxMc8uVGL26RzGXKy8zPN/3JSocNT1fW4JHxN7bxaQZyeR1sUM=
-----END CERTIFICATE-----
```


- Include how the agent connects to the Velociraptor server and how you verify that telemetry is successfully collected.
  Goal: Demonstrate understanding of setup and operational readiness of the EDR environment.

1) CLIENT CONFIGURATION GENERATION

```
[ababil@localhost velociraptor]$ sudo ./velociraptor-v0.76.1-linux-amd64  --config ~/Downloads
/velociraptor/server.config.yaml config client > ~/Downloads/velociraptor/client.config.yaml
[sudo] password for ababil:
[ERROR] 2026-04-13T09:07:51Z ClientInfo Manager: SaveSnapshot:
[ababil@localhost velociraptor]$ ls
client.config.yaml  server.config.yaml  velociraptor-v0.76.1-linux-amd64
[ababil@localhost velociraptor]$ sudo vim client.config.yaml
[ababil@localhost velociraptor]$
```

2) Copy Config to Clients

on Ubuntu (192.168.8.103)

```
[ababil@localhost velociraptor]$ scp client.config.yaml   ababil@192.168.8.103:/home/ababil/Do
wnloads/
The authenticity of host '192.168.8.103 (192.168.8.103)' can't be established.
ED25519 key fingerprint is SHA256:aiqtTwXKb4qofHLpOEmjTKBTP2yAiKRO1kFNn91De1M.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:5: 192.168.1.10
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.8.103' (ED25519) to the list of known hosts.
ababil@192.168.8.103's password:
client.config.yaml                                    100% 2691    795.6KB/s   00:00
[ababil@localhost velociraptor]$
```

```
ababil@ubntuserver:~/Downloads/velociraptor$ ls -lha
total 81M
drwxr-xr-x 2 root   root   4.0K Apr 13 10:10 .
drwxr-xr-x 3 ababil root   4.0K Apr 13 10:10 ..
-rw-r--r-- 1 ababil ababil 2.7K Apr 13 10:09 client.config.yaml
-rwxrwxr-x 1 ababil ababil  81M Apr 13 10:01 velociraptor-v0.76.1-l
inux-amd64
```

on Windows 10 (192.168.8.110)

```
[ababil@localhost ~]$ scp ~/Downloads/velociraptor/clie
nt.config.yaml   user@192.168.8.110:/C:/Users/user/velo
ceraptore/
The authenticity of host '192.168.8.110 (192.168.8.110)
' can't be established.
ED25519 key fingerprint is SHA256:JPDPsXjZUEP4tAKXxv3gY
6vENuFRmJmrlUEEGR/kx6M.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[f
ingerprint])? yes
Warning: Permanently added '192.168.8.110' (ED25519) to
 the list of known hosts.
user@192.168.8.110's password:
dest open("/C:/Users/user/veloceraptore/"): No such fil
e or directory
failed to upload file /home/ababil/Downloads/velocirapt
or/client.config.yaml to /C:/Users/user/veloceraptore/
[ababil@localhost ~]$ scp ~/Downloads/velociraptor/clie
nt.config.yaml   user@192.168.8.110:/C:/Users/user/velo
ciraptor/
user@192.168.8.110's password:
client.config.yaml   100% 2691   207.9KB/s   00:00
[ababil@localhost ~]$
```

```
PS C:\Users\user\velociraptor> ls


    Directory: C:\Users\user\velociraptor


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        20/04/2026     18:40           2691 client.config.yaml
-a----        20/04/2026     18:24       69811192 velociraptor-v0.76.1-windows-amd64.exe
```

3) Deployment Agent on Clients

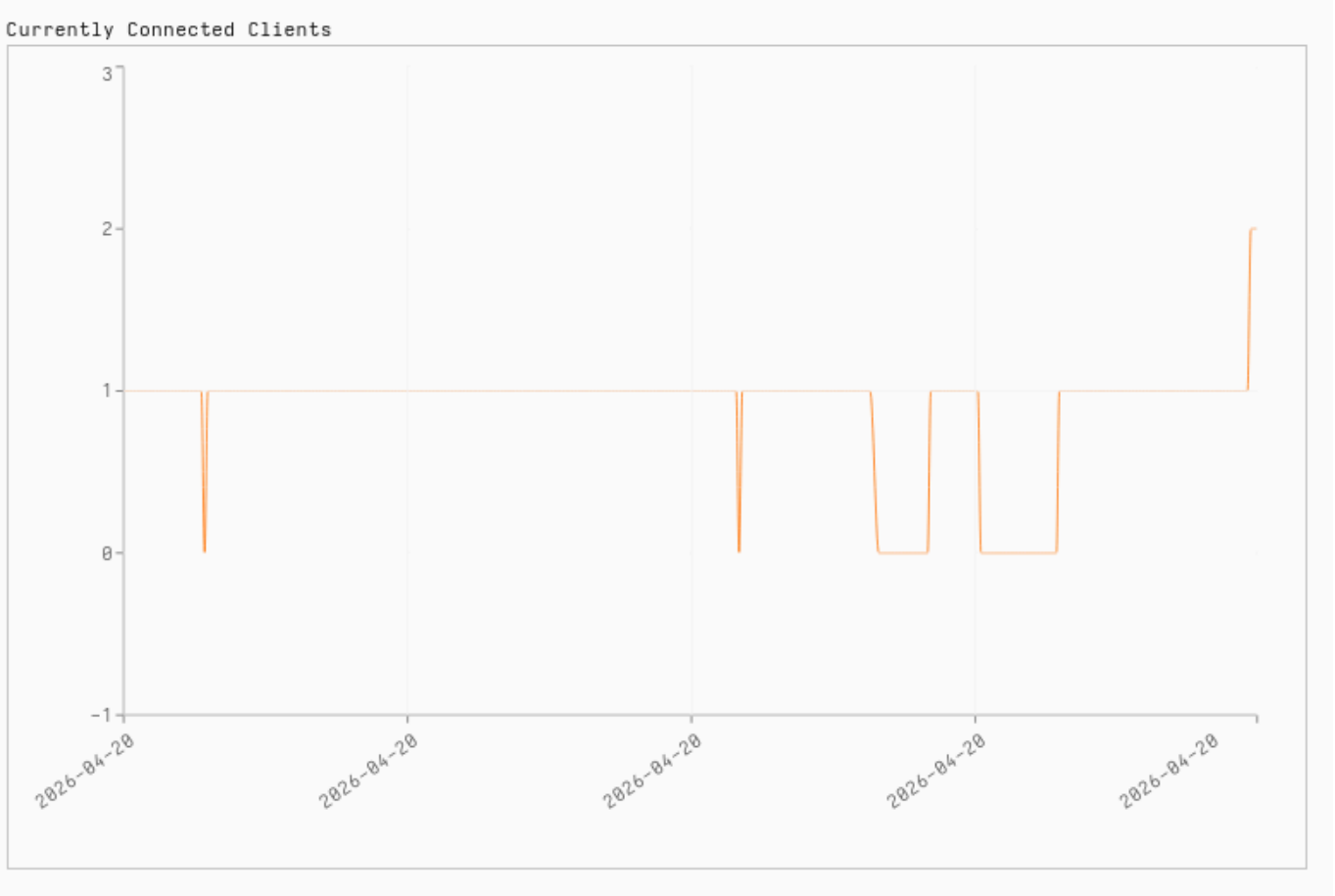
on Ubuntu (192.168.8.103)

```
ababil@ubntuserver:~/Downloads/velociraptor$ chmod +x v
elociraptor-v0.76.1-linux-amd64
ababil@ubntuserver:~/Downloads/velociraptor$ sudo cp ve
lociraptor-v0.76.1-linux-amd64  /usr/local/bin/velocira
ptor
ababil@ubntuserver:~/Downloads/velociraptor$ sudo ./vel
ociraptor-v0.76.1-linux-amd64 --config client.config.ya
ml client -v
[INFO] 2026-04-20T21:30:08Z  _    __     __     _
                  __
[INFO] 2026-04-20T21:30:08Z | |  / /__  / /___  _____(_
)_______ _____  / /_____  _____
[INFO] 2026-04-20T21:30:08Z | | / / _ \/ / __ \/ ___/ /
 ___/ __ `/ __ \/ __/ __ \/ ___/
[INFO] 2026-04-20T21:30:08Z | |/ /  __/ / /_/ / /__/ /
/  / /_/ / /_/ / /_/ /_/ / /
[INFO] 2026-04-20T21:30:08Z |___/\___/_/\____/\___/_/_/
   \__,_/ .___/\__/\____/_/
[INFO] 2026-04-20T21:30:08Z
       /_/
[INFO] 2026-04-20T21:30:08Z Digging deeper!
      https://www.velocidex.com
[INFO] 2026-04-20T21:30:08Z This is Velociraptor 0.76.1
 built on 2026-03-25T16:38:27Z (3e98857b5)
```

on Windows 10 (192.168.8.110)

```
PS C:\Users\user> cd .\velociraptor\
PS C:\Users\user\velociraptor> .\velociraptor-v0.76.1-w
indows-amd64.exe --config client.config.yaml client -v
[INFO] 2026-04-20T23:04:06+01:00  _    __     __
    _                    __
[INFO] 2026-04-20T23:04:06+01:00 | |  / /__  / /___  __
___(_)_______ _____  / /_____  _____
[INFO] 2026-04-20T23:04:06+01:00 | | / / _ \/ / __ \/ _
__/ / ___/ __ `/ __ \/ __/ __ \/ ___/
[INFO] 2026-04-20T23:04:06+01:00 | |/ /  __/ / /_/ / /_
_/ / /  / /_/ / /_/ / /_/ /_/ / /
[INFO] 2026-04-20T23:04:06+01:00 |___/\___/_/\____/\___
/_/_/   \__,_/ .___/\__/\____/_/
[INFO] 2026-04-20T23:04:06+01:00
            /_/
[INFO] 2026-04-20T23:04:06+01:00 Digging deeper!
           https://www.velocidex.com
[INFO] 2026-04-20T23:04:06+01:00 This is Velociraptor 0
.76.1 built on 2026-03-25T16:40:56Z (3e98857b5)
[INFO] 2026-04-20T23:04:06+01:00 Loading config from fi
le client.config.yaml
[INFO] 2026-04-20T23:04:06+01:00 Writeback Manager: Una
ble to read writeback (open C:\Program Files\Velocirapt
or\velociraptor.writeback.yaml: The system cannot find
the path specified.) - will reset
[ERROR] 2026-04-20T23:04:06+01:00 Writeback Manager:  W
riteL1: Writeback WriteFile to C:\Program Files\Velocir
```

Show all runing Agent (2)


Currently Connected Clients
3
2
1
0
-1
2026-04-20
2026-04-20
2026-04-20
2026-04-20
2026-04-20

Activities
Firefox
Apr 20 23:40
Velociraptor Response an
localhost:8889/app/index.html?org_id=root#/search/all
80%
all
admin <root>
0-3/3
10
all
Client ID
Hostname
FQDN
OS Version
Labels
C.cd14ffe02745e5d1
DESKTOP-CUTQRHP
DESKTOP-CUTQRHP
Microsoft Windows 10 Home22H2
C.d7491b0f532f5d07
ubntuserver
ubntuserver
ubuntu20.04
C.da03f65d43f8b7bb
localhost.localdomain
localhost
redhat9.7

Scenario

Your SOC has received alerts indicating **suspicious activity on a workstation**. Your mission is to

1. Observe alerts from Wazuh
2. Investigate the endpoint using Velociraptor
3. Determine what the attacker did

Phase 1 – Attack Simulation (first attack: brute force login)

From **Kali Linux**, simulate suspicious activity. Perform multiple failed login attempts against the Windows machine.

Expected observations

- Authentication failures
- Possible alert from Wazuh

You must identify

- target username
- source IP
- number of attempts

MITRE mapping: **T1110 – Brute Force**

Phase 2 – Suspicious PowerShell Execution

On the Windows endpoint, run a suspicious PowerShell command. For example:

```
powershell -EncodedCommand ...
```

You must investigate using **Velociraptor artifacts**

- process list
- command-line history
- PowerShell logs

MITRE mapping: **T1059 – Command and Scripting Interpreter**

Phase 3 – Privilege Escalation

Create a new user and add it to the **Administrators group**. Then, you must investigate

- user accounts
- privilege changes
- related events

Velociraptor artifacts to analyze

- Windows event logs
- user accounts list

MITRE mapping: **T1098 – Account Manipulation**

Phase 4 – Endpoint Investigation

You must perform a **live investigation using Velociraptor**.

Example artifacts to collect

- running processes (`pslist`)
- network connections (`netstat`)
- user sessions
- event logs
- suspicious files

You must determine

- Which process was malicious
- What commands were executed
- When the attack started

Advanced Task – Attack Timeline

You must reconstruct the **attack timeline**. For example:

Time	Event
02:05 PM	Brute force attempts
02:08 PM	Successful login
02:10 PM	PowerShell execution
02:12 PM	Privilege escalation

This step simulates **SOC incident analysis**.

Required Report Structure

For each phase, you must include

1. Attack description
2. Steps used to simulate the attack
3. Alerts generated by Wazuh
4. Evidence collected using Velociraptor
5. Analysis of the attacker's behavior
6. MITRE ATT&CK mapping

7. Security impact

Include screenshots of

- alerts
- Velociraptor queries
- investigation artifacts